STEP Write Ups

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Assignment 1

Let $a, b \in \mathbb{R}\backslash\{0\}$ and $c \in \mathbb{R}$.

(i) Prove that, if a and b are either both positive or both negative, then the equation

$$\frac{x}{x-a} + \frac{x}{x-b} = 1$$

has two distinct real solutions.

Proof. Assume that $x \neq a$ and $x \neq b$ The equation can be rearranged by multiplying by $x - a$ and $x - b$ to get

$$x(x-a) + x(x-b) = (x-a)(x-b),$$

which can be simplified to

$$x^2 - ab = 0.$$

In either case, we know that $ab > 0$ and so the real solutions of this equation are $x = \pm\sqrt{ab}$. □

(ii) Prove that, if $c \neq 1$, the equation

$$\frac{x}{x-a} + \frac{x}{x-b} = 1 + c$$

has exactly one real solution if

$$c^2 = -\frac{4ab}{(a-b)^2}.$$

Prove that this can be written

$$c^2 = 1 - \left(\frac{a+b}{a-b}\right)^2$$

and deduce that it can only hold if $0 < c^2 \leq 1$.

Proof. Similarly, the equation can be reduced to

$$x^2 - ab = c(x-a)(x-b),$$

or alternatively,

$$(1-c)x^2 + c(a+b)x - (c+1)ab = 0.$$

To find the case where this equation has one real root, set the discriminant to zero, i.e.

$$c^2(a+b)^2 - 4(1-c)(c+1)(-ab) = 0.$$

Rearranging this, the result becomes

$$c^2 = -\frac{4ab}{(a-b)^2}.$$

Since $-4ab = a^2 - 2ab + b^2 - (a^2 + 2ab + b^2) = (a-b)^2 - (a+b)^2$, the equation for c^2 becomes

$$\begin{aligned} c^2 &= -\frac{4ab}{(a-b)^2} \\ &= \frac{(a-b)^2 - (a+b)^2}{(a-b)^2} \\ &= 1 - \left(\frac{a+b}{a-b}\right)^2. \end{aligned}$$

Since c^2 is a real number, that necessarily means that it is positive. However, for c^2 to be zero, $-4ab$ has to be zero, which requires one of a, b to be zero, which goes against the hypothesis of the question. Thus, $0 < c^2 \leq 1$. □

Assignment 2

(i) Find the greatest and least values of $bx + a$ for $-10 \leq x \leq 10$.

Proof. For $b = 0$, we are left with just a as the greatest and least values. For $b > 0$, the greatest value occurs at $x = 10$, which is $10b + a$. The least value occurs at $x = -10$, which is $-10b + a$. For $b < 0$, the greatest value occurs at $x = -10$, which is $-10b + a$. The least value occurs at $x = 10$, which is $10b + a$. $\square$

(ii) Find the greatest and least values of $cx^2 + bx + a$, where $c \geq 0$ for $-10 \leq x \leq 10$.

Proof. For $c = 0$, we have the case of the linear equation above. For $c > 0$, we complete the square to get

$$\left(x + \frac{b}{2c}\right)^2 + \frac{a}{c} - \frac{b^2}{4c^2}.$$

We can use differentiation to find that the minimum of this expression exists at $x = \frac{-b}{2c}$. For the greatest value, we consider different values of b. When $b = 0$, we have the same greatest value at $x = 10, x = -10$ of $100c + a$. When $b < 0$, then the greatest value is at $x = 10$, with $100c + a$. When $b > 0$, then the greatest value is at $x = -10$, with $100c + a$. These last two conditions are because we are only considering the expression over a closed interval. $\square$

Assignment 3

(i) Prove that

$$\int_0^a \sqrt{\lfloor x \rfloor}\, \mathrm{d}x = \sum_{r=0}^{a-1} \sqrt{r}.$$

when a is a positive integer.

Proof. Using the additivity, the integral becomes

$$\begin{aligned}\int_0^a \sqrt{\lfloor x \rfloor}\, \mathrm{d}x &= \int_0^1 \sqrt{\lfloor x \rfloor}\, \mathrm{d}x + \int_1^2 \sqrt{\lfloor x \rfloor}\, \mathrm{d}x + \ldots \int_{a-1}^a \sqrt{\lfloor x \rfloor}\, \mathrm{d}x \\ &= \sum_{r=0}^{a-1} \left(\int_r^{r+1} \sqrt{\lfloor x \rfloor}\, \mathrm{d}x \right)\end{aligned}$$

On the interval $[r, r+1)$, $\lfloor x \rfloor = r$, so $\sqrt{\lfloor x \rfloor} = \sqrt{r}$ for all x on the interval. Thus, it follows that

$$\int_0^a \sqrt{\lfloor x \rfloor}\, \mathrm{d}x = \sum_{r=0}^{a-1} \sqrt{r}.$$

$\square$

(ii) Prove that $\int_0^a 2^{\lfloor x \rfloor}\, \mathrm{d}x = 2^a - 1$.

Proof. Using a similar argument as previous, we aim to transform this floor function into a summation. On the interval $[r, r+1)$, we have that $\lfloor x \rfloor = r$ and thus $2^{\lfloor x \rfloor} = 2^r$. The integral becomes

$$\begin{aligned}\int_0^a 2^{\lfloor x \rfloor}\, \mathrm{d}x &= \sum_{r=0}^{a-1} \left(\int_r^{r+1} 2^{\lfloor x \rfloor}\, \mathrm{d}x \right) \\ &= \sum_{r=0}^{a-1} 2^r.\end{aligned}$$

Since this is a geometric progression, it can be easily evaluated as

$$\frac{1(1 - 2^a)}{1 - 2} = 2^a - 1.$$

$\square$

(iii) Determine an expression for $\int_0^a 2^{\lfloor x \rfloor}\,\mathrm{d}x$ when a is positive but not an integer.

Proof. Using the floor function, the integral can be split into

$$\int_0^a 2^{\lfloor x \rfloor}\,\mathrm{d}x = \int_0^{\lfloor a \rfloor} 2^{\lfloor x \rfloor}\,\mathrm{d}x + \int_{\lfloor a \rfloor}^a 2^{\lfloor x \rfloor}\,\mathrm{d}x.$$

We derived an expression for the first part of this sum in the previous question. We still have that the value at any point in $[\lfloor a \rfloor, a)$ will be 2^a, but we multiply by $(a - \lfloor a \rfloor)$ in order to determine the fractional part. Thus, the expression is

$$\int_0^a 2^{\lfloor x \rfloor}\,\mathrm{d}x = 2^a - 1 + (a - \lfloor a \rfloor)2^a.$$

□

Assignment 4

(i) Find the real values of x for which $x^3 - 4x^2 - x + 4 \geq 0$.

Proof. Guessing that $x = 4$ is a root (which it is), we are able to find that the other two roots are $x = \pm 1$. Sketching the graph and using the knowledge of behaviour of cubics as $x \to \pm\infty$, we know that the expression is greater than 0 is when $-1 \leq x \leq 1$ or $x \geq 4$. □